

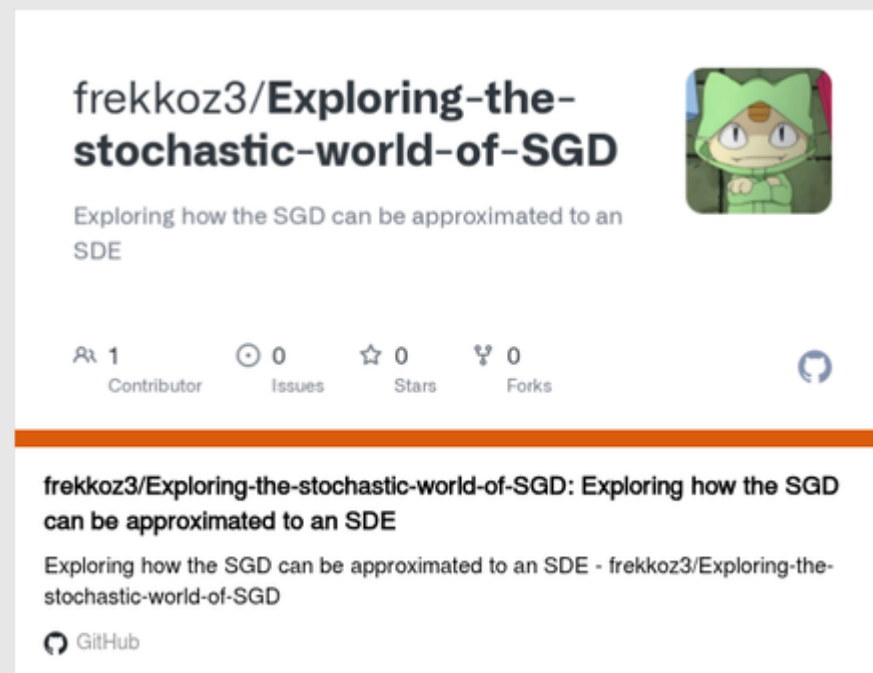
ABOUT SGD

a theoretical and practical journey about this incredible optimization algorithm

setup

Project Overview

The aim for this project is to analyze how the Stochastic Gradient Descent is linked to the world of Stochastic Differential Equation, approaching it from multiple direction, which are both theoretical and practical. For that, an educational series of notebook has been proposed.



Theory of SGD as SDE

Visualization of SGD

Other Stochastic Techniques

theory

Problem Formulation

Given paired data sample

$$X = (x_1, \dots, x_N) \quad Y = (y_1, \dots, y_N)$$

and a parametric model

$$f(X, \theta)$$

we define the Loss Function as

$$L(\theta) = \frac{1}{N} \sum_{i=1}^N \ell_i(\theta)$$

the aim is to find

$$\theta^* = \arg \min_{\theta} L(\theta)$$

theory

GD and SGD

Gradient Descent

$$\theta_{k+1} = \theta_k - \eta \nabla L(\theta_k)$$

$$\nabla L_{B_k}(\theta) = \frac{1}{N} \sum_{i=1}^N \nabla \ell_i(\theta)$$

Stochastic Gradient Descent

$$\theta_{k+1} = \theta_k - \eta \nabla \tilde{L}_{B_k}(\theta_k)$$

$$\nabla \tilde{L}_{B_k}(\theta) = \frac{1}{|B_k|} \sum_{i \in B_k} \nabla \ell_i(\theta)$$

uniformly sampled substed of data

theory

Standard SDE Baseline

In the literature, the standard approximation of the SGD as SDE is the following (obtained via moment matching):

$$d\theta_t = -\nabla L(\theta_t)dt + \sqrt{\eta}\sigma dW_t$$

This baseline treats the discrete optimization path as a discretization of an Overdamped Langevin Diffusion process, relying on assumptions of **isotropic, state-independent Gaussian noise**.

theory

Long Term Perspective

Cheng et al in “Gradient Descent and Langevin Process”

The real complex nature of the noise motivates studying Langevin-like methods with **non-Gaussian** and **state-dependent** noise, where the potential function can be **non-convex**.

$$w_{(k+1)\delta} = w_{k\delta} - \eta \nabla U(w_{k\delta}) + \sqrt{\eta} \xi(w_{k\delta}, b_k) \quad (1)$$

non-gaussian, state-dependent noise

EULER MAYURANA DISCRETIZATION OF (3)

$$dy_t = -\nabla U(y_{k\delta})dt + M(y_{k\delta})dW_t \quad (2)$$

TARGET SDE

$$dx = -\nabla U dt + M(x_t)dW_t \quad (3)$$

theory

Theorems key Insights

The paper provide quantitative, finite-time bounds on how well (1) and (2) track the invariant distribution p^* of (3), without Gaussian or convexity assumptions. From the two theorems provided in the paper there are two main takeaways:

- **Primacy of the noise covariance:** over long horizons only $M(x)M(x)^T$ matters. Higher-order moments average out, justifying Gaussian-SDE models
- **De-biasing small mini-batches:** theorem 2 removes the large-batch CLT requirement, proving the SDE governs even small-batch dynamics

theory

Local High order Correction

Li et al. "Stochastic Modified Equation and Adaptive Stochastic Gradient Algorithms"

Standard SDE approximations rely on $\eta \rightarrow 0$ and isotropic Gaussian noise. In practice, discrete steps leave an **unaddressed remainder of order $O(\eta)$** in the Taylor expansion.

The paradigm shift: instead of matching trajectories, find the SDE that matches the **probability distribution of the path**.

theory

Modified Equations

Adapted from deterministic numerical analysis. Instead of asking "how well does discrete GD approximate the ODE?", ask: "**what modified ODE does GD follow exactly?**" (up to a certain order of accuracy)

$$\dot{x} = -\nabla \left(f(x) + \frac{\eta}{4} \|\nabla f(x)\|^2 \right) + \mathcal{O}(\eta^2)$$

1st order Taylor Expansion of the
continuous path interpolating GD

- Discrete GD does not minimize $f(x)$, it minimizes a regularized landscape $\tilde{f}(x) = f(x) + (\eta/4) \|\nabla f\|^2$
- Regions with steep gradients are penalized, implicit regularization emerges naturally from the discrete step size
- This deterministic insight extends directly to the stochastic domain via Stochastic Modified Equations (SME)

theory

Stochastic ME

In the paper *Li et al* show these two main results

ORDER 1 WEAK APPROXIMATION OF THE SGD

$$dX_t = -\nabla f(X_t)dt + \sqrt{\eta}\Sigma(X_t)^{\frac{1}{2}}dW_t$$

Standard baseline,
state-dependent noise but standard drift

ORDER 2 WEAK APPROXIMATION OF THE SGD

$$dX_t = -\nabla \left(f(X_t) + \frac{\eta}{4} \|\nabla f(X_t)\|^2 \right) dt + \sqrt{\eta}\Sigma(X_t)^{\frac{1}{2}}dW_t$$

drift corrected diffusion unchanged : higher-order accuracy
comes purely from the drift term

$$\Sigma(x) = \frac{1}{n} \sum_{i=1}^n (\nabla f(x) - \nabla f_i(x))(\nabla f(x) - \nabla f_i(x))^T$$

exact **gradient covariance matrix**

theory

Algorithmic Design

The main contribution of the paper is showing that, given those two theoretical results one can try to treat the SGD as a diffusion process and uses Optimal Control tools (or whatever one known) in order to design **novel optimization algorithms**.

Li et al. propose **cSGD**, a variant of SGD where they introduced a **throttle parameter** $u(t) \in [0, 1]$ that **scales the learning rate** automatically.

- $u(t) = 1$: optimizer at full speed
- $u(t) \rightarrow 0$: optimizer slows down near noisy regions
- cSGD runs a lightweight EMA regression to estimate local curvature and noise level
- Computes the optimal throttle per weight, mimicking the ideal theoretical control policy

Visualizing SGD

Notebook 2 moves from theory to intuition. In the first part we visualize how **GD** and the **2nd-order SME proxy** for SGD traverse toy loss landscapes. In the second part we visualize how **GD against SGD** moves in the parameter space while optimizing some models.

SME PROXY UPDATE RULE

$$X_{t+1} = X_t - \eta \nabla \tilde{f}(X_t) + \sqrt{\eta} \xi_t$$

isotropic gaussian noise

$$\tilde{f}(X_t) = f(X_t) + \frac{\eta}{4} \|\nabla f(X_t)\|^2$$

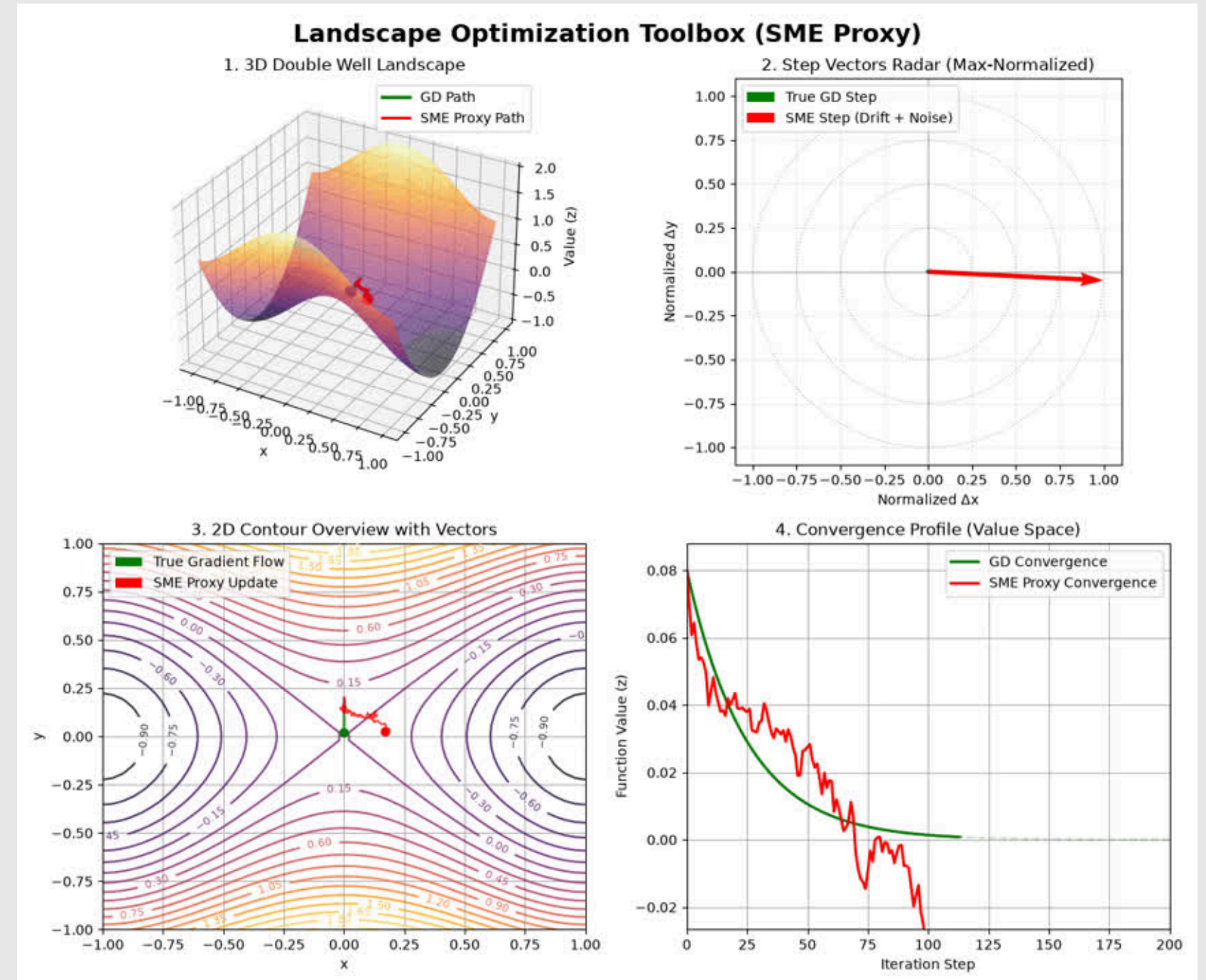
regularized landscape

visualization

Double Well

$$z = x^4 - 2x^2 + 2y^2$$

two symmetric minimum,
one saddle point in (0, 0)

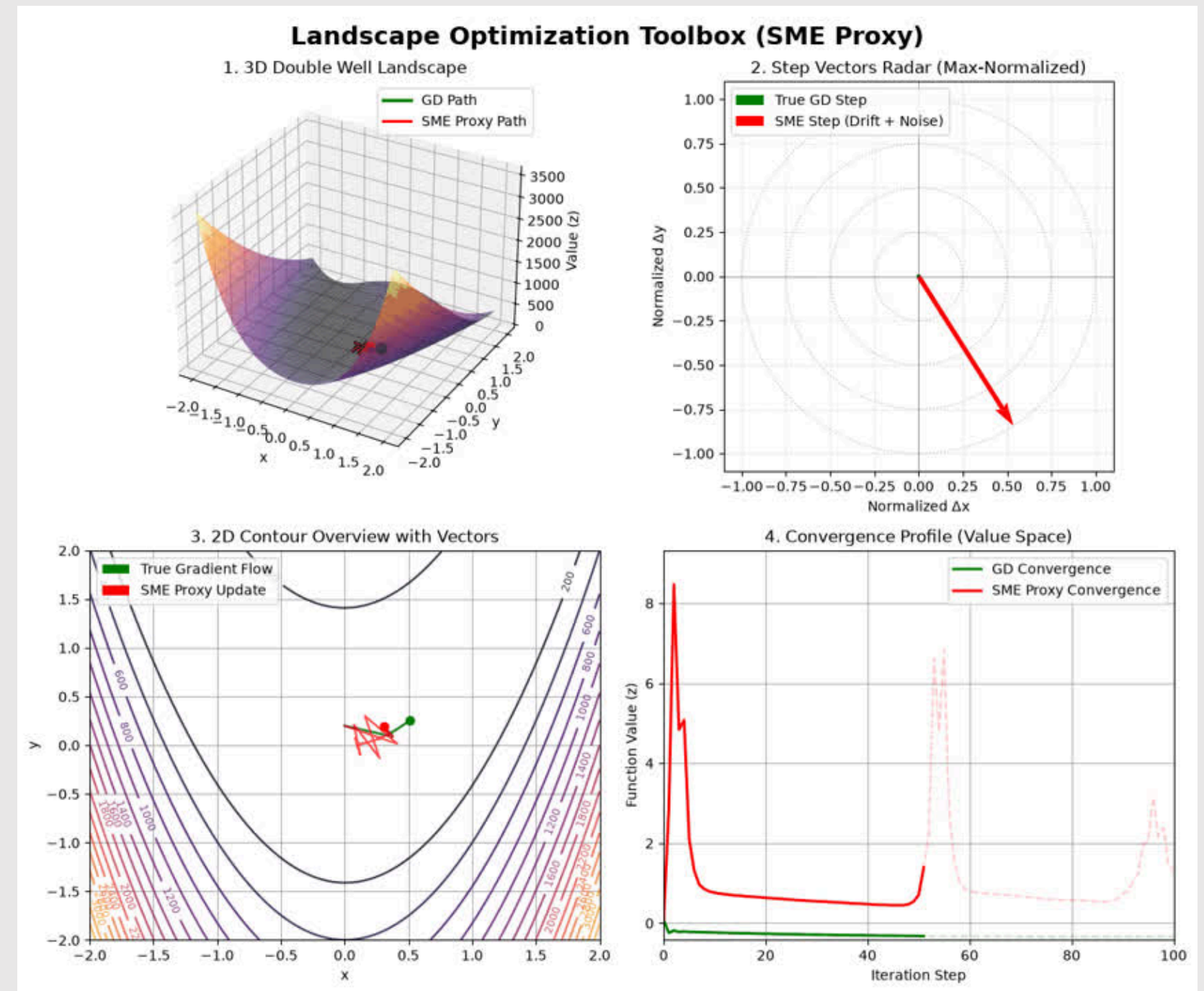


visualization

Rosenbrock

$$z = (1 - x)^2 + 100(y - x^2)^2$$

a minimum located in a very flat, narrow, parabolic trough



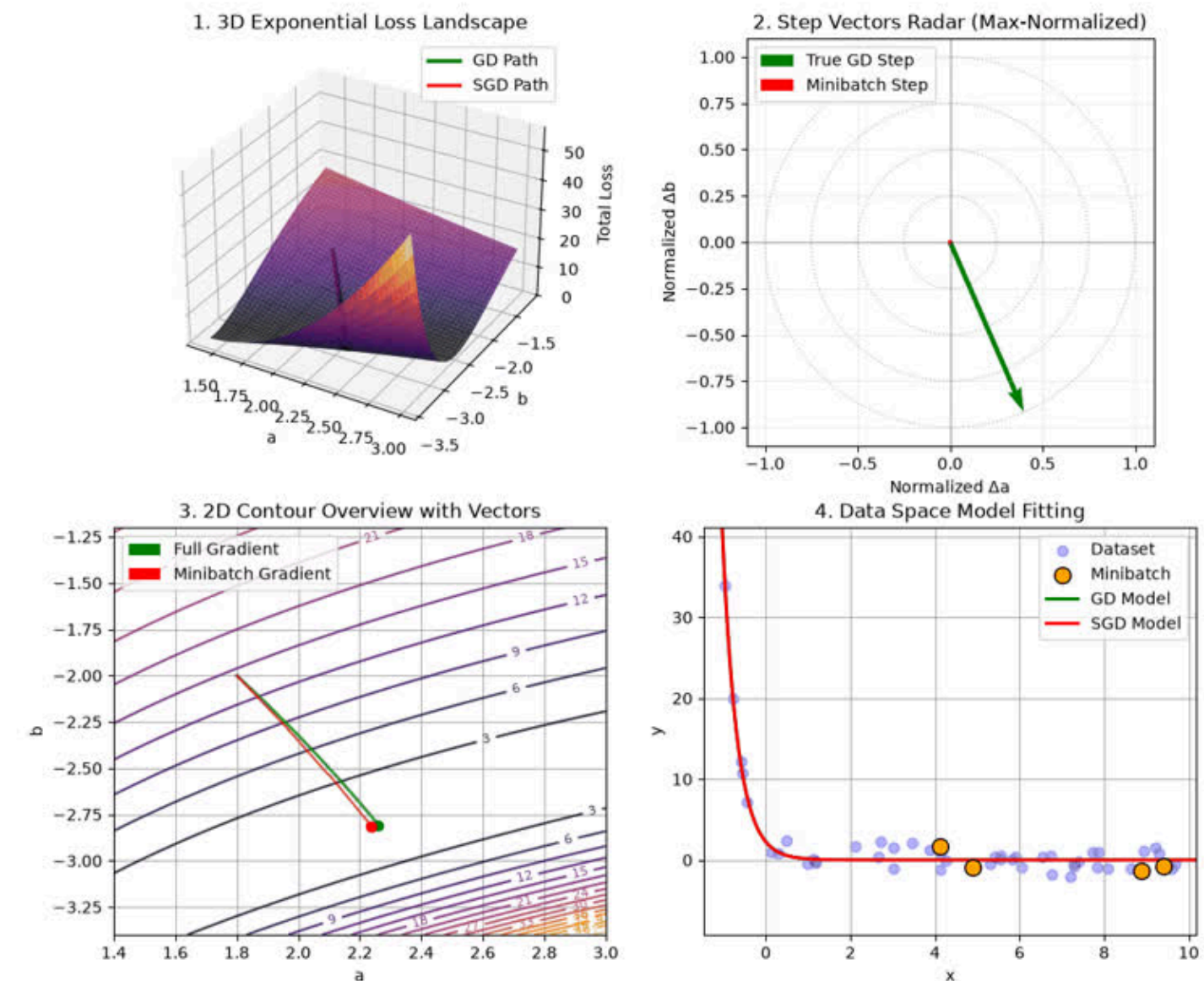
visualization

Exponential Model

$$f(a, b, x) = ae^{bx}$$

convex landscape with
a minimum sitting on flat rift

Optimization Visualization Toolbox



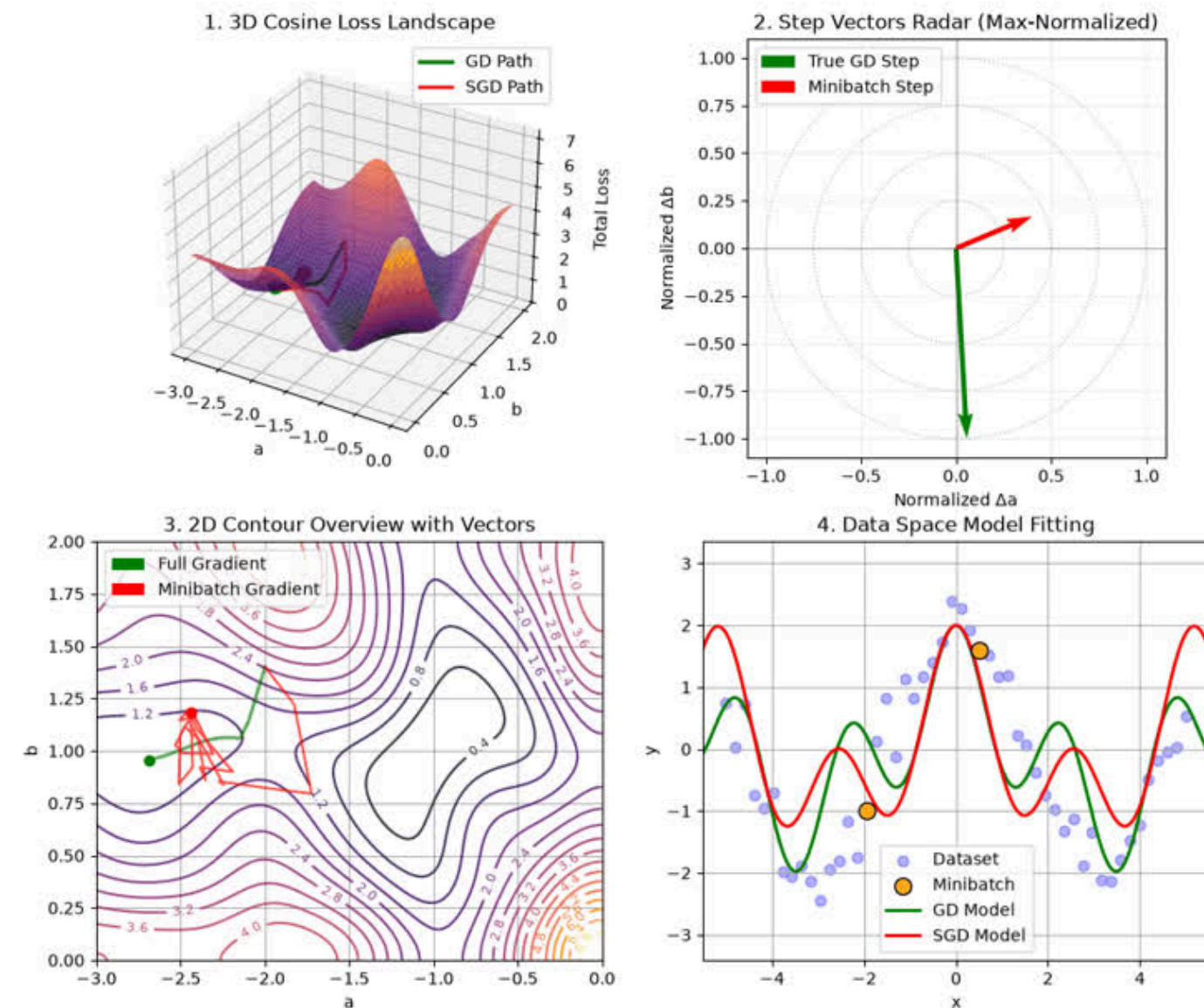
visualization

Cosine Model

$$f(a, b, x) = \cos(ax) + \cos(bx)$$

landscape with multiple local minimum
and a lot of symmetries

Optimization Visualization Toolbox



comparison

Other Opt Techniques

The last notebook compares the SGD against other stochastic optimization techniques.

It keeps a focus on highlighting the **source and structure of randomness** in each method, not just optimization performance.

- SGD
- Momentum SGD
- Simulated Annealing
- Particle Swarm Optimization
- SPHF

comparison

Momentum SGD

The standard Momentum SGD update schema is a **coupled schema** of **position** and **velocity**.

$$v_{t+1} = \beta v_t + (1 - \beta) \nabla f(x_t)$$

$$x_{t+1} = x_t - \eta v_{t+1}$$

Velocity is indeed an exponentially weighted moving average of past gradients and the effective memory horizon is

$$\tau \sim \frac{1}{1 - \beta}$$

comparison

Simulated Annealing

Simulated Annealing does not use gradient information. Instead, it relies solely on **function evaluations**. At iteration t , a candidate point x' is generated from the current point $x(t)$ through a **random perturbation**. The candidate is accepted according to the **Metropolis criterion**

$$P(\text{accept}) = \begin{cases} 1, & f(x') < f(x_t), \\ \exp\left(-\frac{f(x') - f(x_t)}{T_t}\right), & f(x') \geq f(x_t), \end{cases}$$

In the meanwhile the temperature follow a **scheduler** that **decrease** the **possibility of** random jump toward **worst solutions**.

comparison PSO

Particle Swarm Optimization (PSO) belongs to the family of bio-inspired optimization methods known as **swarm intelligence**. The individual particles are simple agents with limited capabilities; complex and adaptive behaviour emerges from their **collective interactions**.

$$v_i^{t+1} = \omega v_i^t + c_1 r_1 (p_i - x_i^t) + c_2 r_2 (g - x_i^t)$$

$$r_1, r_2 \sim U(0, 1)$$

$$x_i^{t+1} = x_i^t + v_i^{t+1}$$

comparison SPHF

To better isolate the stochastic mechanisms underlying PSO, we introduce a toy algorithm that we call **Sad Particle Hallucinating Friends (SPHF)** algorithm.

$$v_{t+1} = wv_t + c_1 r_{1,t}(p_t - x_t) + c_2 r_{2,t}(g_t - x_t)$$
$$r_1, r_2 \sim U(0, 1)$$

$$x_{t+1} = x_t + v_t$$

HALLUCINATION + MEMORY

$$g_t = \arg \min_{y \in y_t^{(1)}, \dots, y_t^{(N)}, g_{t-1}} f(y)$$

$$y_t^{(1)}, \dots, y_t^{(N)} \sim q_t$$

Makes the stochastic mechanism easier to analyze than full PSO. **Two clean noise sources:** coefficients r_1, r_2 and hallucination sampling.

comparison

Summary Table

Method	Randomness Source	Search Scope	Gradient Dependence
SGD	Data mini-batching	Local	Yes
Momentum SGD	Accumulated Data mini-batching	Local	Yes
Simulated Annealing	Proposal Distribution Step	Local → Global	No
PSO	Random Coefficients	Local / Global	No
SPHF	Coefficients + Friends Sampling	Local / Global	No

comparison **Empirical Taste**

To finish the project we developed a simple **empirical comparison** of the various method on different types of landscapes (each of whom with **peculiar and unique type of properties**).

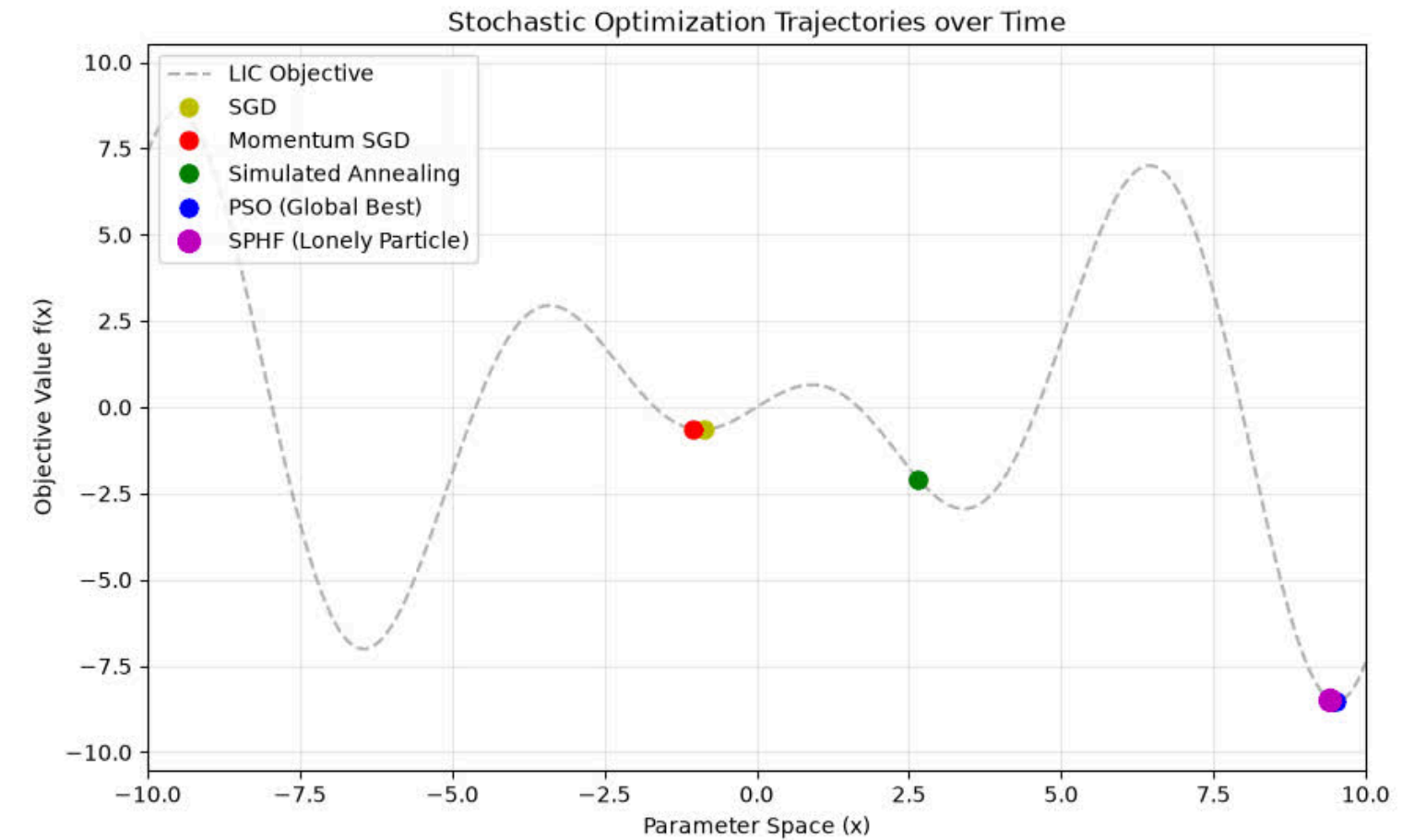
- 1D Bounded Linear Increasing Cosine
- 2D Bounded Rastrigin
- HD Bounded Rastrigin
- HD Bounded Elipsoid

comparison

Linear Increasing Cosine

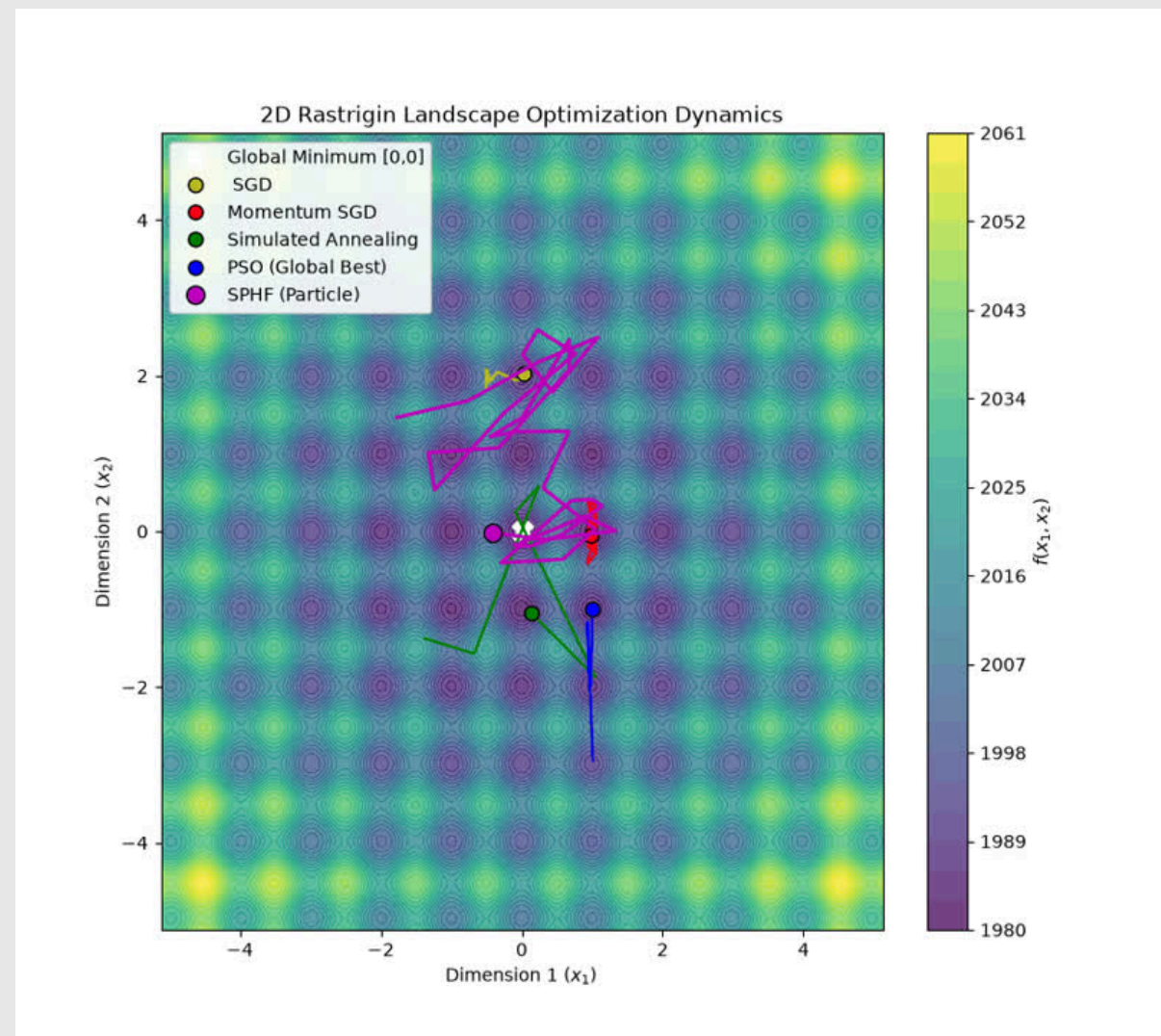
$$f(x) = x \cos(x) + 0.1x$$

simple yet stresses perfectly the flaws
of gradient-based methods



comparison 2D Rastrigin

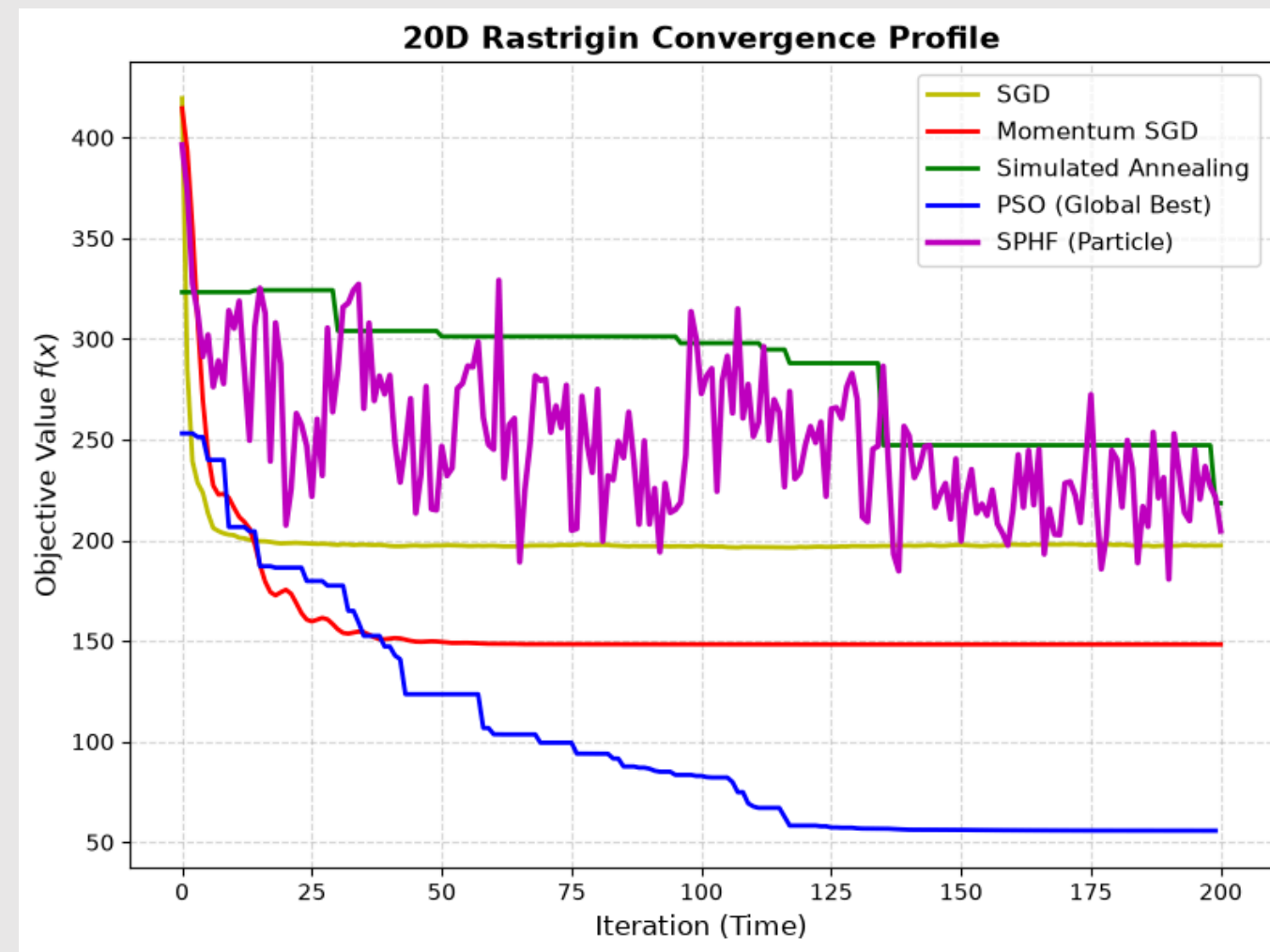
$$f(x, y) = 20 + (x^2 - 10 \cos(2\pi x)) + (y^2 - 10 \cos(2\pi y))$$



highly non-convex with a massive number of regularly distributed local minima

comparison HD Rastrigin

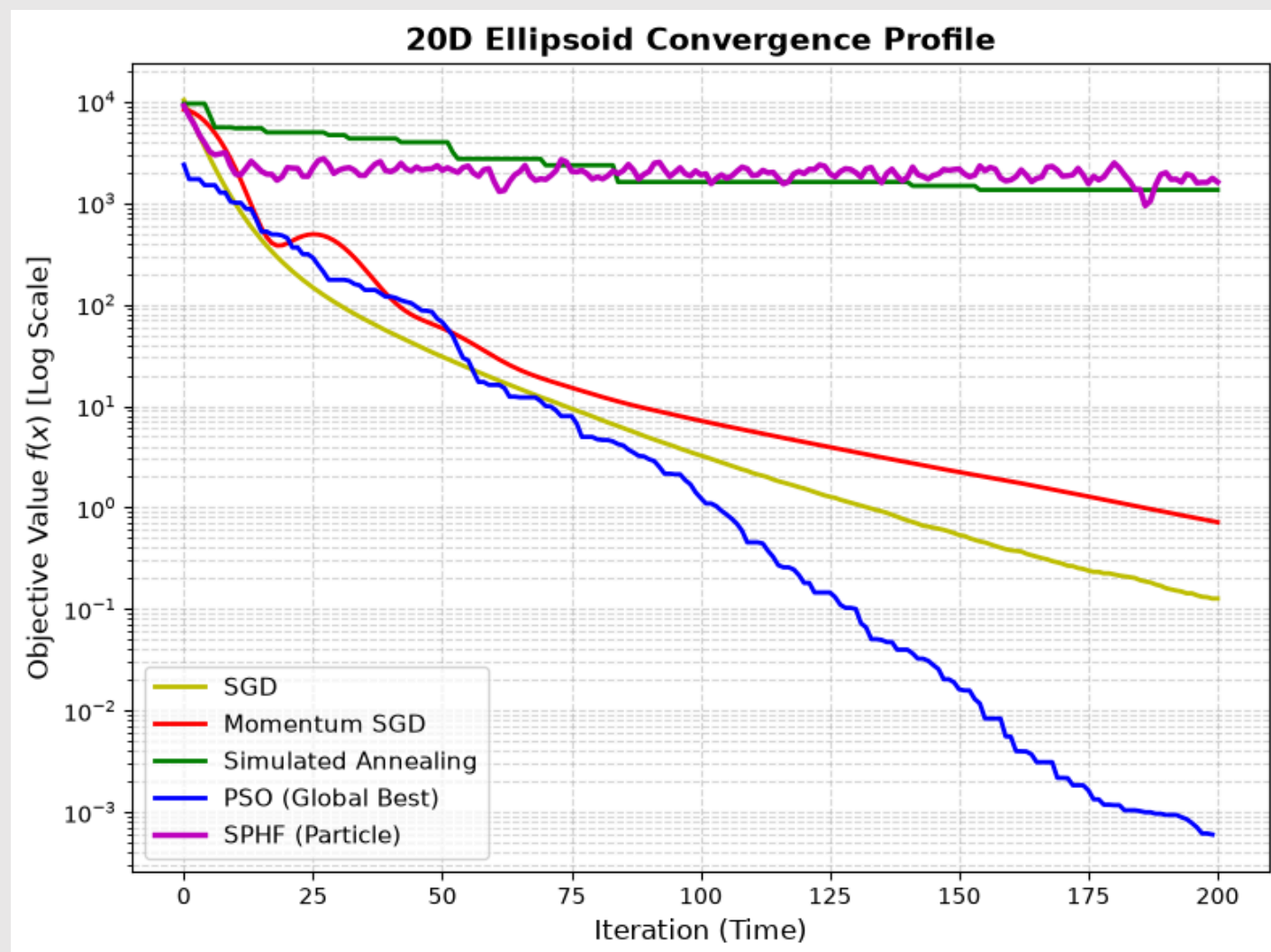
$$f(x) = 10D + \sum_{i=1}^D (x_i^2 - 10\cos(2\pi x_i))$$



highly non-convex with a massive number of
regularly distributed local minima
the real minimum is located in vector 0
with a value of 0

comparison HD Ellipsoid

$$f(\mathbf{x}) = \sum_{i=1}^D i \cdot x_i^2$$



convex landscape with a really
flat valley where the minimum lives

thank you

Francesco Bredariol